

Linear Algebra II (2026 Fall)

Lecture 6 and Lecture 7

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Chapter 5. Diagonalization

- Section 5.4 : Invariant Subspaces and the Cayley-Hamilton Theorem

Section 5.4 Invariant Subspaces and the Cayley-Hamilton Theorem

Remark : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V .

If λ is an eigenvalue of T and $v \in V$ is an eigenvector of T corresponding to λ , then

$$T(cv) = cT(v) = c\lambda v \in \langle v \rangle \text{ for all } cv \in \langle v \rangle.$$

That is, T maps the subspace $\langle v \rangle$ into itself.

Similarly, if E_λ is the eigenspace T corresponding to λ then T maps E_λ into itself.

Definition : Let $T : V \rightarrow V$ be a linear transformation of a vector space V .

A subspace W of V is called a T -invariant subspace of V if $T(W) \subseteq W$,

or equivalently, if $T(v) \in W$ for all $v \in W$.

Example 1. : Let $T : V \rightarrow V$ be as above. Then the following subspaces of V are T -invariant:

- $\{O_V\}$ and V
- $R(T) = \{T(v) \mid v \in V\}$ and $N(T) = \{v \in V \mid T(v) = O_V\}$
- E_λ , for any eigenvalue λ of T

Example 2. : Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation defined by

$$T(a,b,c) = (a+b, b+c, 0).$$

(1) The xy -plane $\{ (x,y,0) \mid x,y \in \mathbb{R} \}$ is a T -invariant subspace of V .

Indeed, $R(T)$ is equal to the xy -plane.

(2) The x -axis $\{ (x,0,0) \mid x \in \mathbb{R} \}$ is a T -invariant subspace of V .

Since $[T]_{\beta} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, we get $\det([T]_{\beta} - tI_3) = \det \begin{pmatrix} 1-t & 1 & 0 \\ 0 & 1-t & 1 \\ 0 & 0 & -t \end{pmatrix} = -t(1-t)^2 = -t(t-1)^2$.

$(t = 1)$

If $([T]_{\beta} - I) \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{pmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, then $y = z = 0$ and hence $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$.

Thus the x -axis is the eigenspace of T corresponding to the eigenvalue $t = 1$.

Definition and Remark : Let $T : V \rightarrow V$ be a linear transformation of a vector space V .

(1) For a nonzero vector $x \in V$, the subspace

$$W := \text{span}(\{x, T(x), T^2(x), \dots\}) = \text{span}(\{T^\ell(x) \mid \ell \geq 1\})$$

is called **the T -cyclic subspace of V generated by x** .

(2) W is the smallest T -invariant subspace of V containing x .

That is, W is a T -invariant subspace of V such that

any T -invariant subspace of V containing x must also contain W .

(3) Suppose that $T^{\ell+1}(v) \in \text{span}(\{v, T(v), \dots, T^\ell(v)\})$.

Then $W = \text{span}(\{v, T(v), \dots, T^\ell(v)\})$.

Example 3. : In Example 2, if $v = (1,0,0)$ then $T(v) = (1,0,0)$ and $T^2(v) = (1,0,0)$.

Thus the T -cyclic subspace of $\mathbb{R}^3$ generated by v is equal to $\text{span}(v)$.

This proves that the x -axis is a T -invariant subspace of $\mathbb{R}^3$.

Remark. : In Example 2, if $v = (1,1,0)$ then $T(v) = (2,1,0)$ and $T^2(v) = (3,1,0)$.

In particular, $T^2(v) = 2T(v) - v \in \text{span}(v, T(v))$.

Thus the T -cyclic subspace of $\mathbb{R}^3$ generated by v is equal to $\text{span}(v, T(v))$.

Also, one can easily see that $\text{span}(v, T(v))$ is the xy -plane.

This reproves that the xy -plane is a T -invariant subspace of $\mathbb{R}^3$.

Example 4. : Let $T : P(\mathbb{R}) \rightarrow P(\mathbb{R})$ be the linear operator defined by $T(f(x)) = f'(x)$.

(1) For every positive integer n , let W_n be the T -cyclic subspace of $P(\mathbb{R})$ generated by x^n .

Then $W_n = \text{span}(x^n, x^{n-1}, \dots, x, 1) = P_n(\mathbb{R})$.

(2) Let W be the T -cyclic subspace of $P(\mathbb{R})$ generated by $x^3 - 3x$.

Then $W = \text{span}(x^3 - 3x, 3x^2 - 3, 6x, 6) = P_3(\mathbb{R})$.

(3) Let W be the T -cyclic subspace of $P(\mathbb{R})$ generated by a polynomial of degree $n \geq 1$.

Then prove that $W = P_n(\mathbb{R})$.

We apply the T -cyclic subspaces of V in this section to establish the Cayley-Hamilton theorem.

Theorem 5.23 (Cayley-Hamilton) : For a linear operator T on an n -dimensional vector space V , let $f(t) = b_0 + b_1t + \cdots + b_nt^n$ be its characteristic polynomial. Then T satisfies $f(t)$ in the sense that

$$f(T) := b_0Id_V + b_1T + b_2T^2 + \cdots + b_nT^n \text{ is the zero map.}$$

Remark : Suppose that $T : F^n \rightarrow F^n$ is defined by $A \in M_{n \times n}(F)$.

Then the Cayley-Hamilton Theorem says that

$$f(A) := b_0I_n + b_1A + b_2A^2 + \cdots + b_nA^n \text{ is the } n \times n \text{ zero matrix.}$$

Example A :

$$\text{For } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \text{ we have } f(t) = \det \begin{pmatrix} a-t & b \\ c & d-t \end{pmatrix} = t^2 - (a+d)t + (ad-bc).$$

$$\text{Then one can check, by hand, that } f(A) = A^2 - (a+d)A + (ad-bc)I_2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Let $T : V \rightarrow V$ be a linear operator of a vector space V .

The existence of a T -invariant subspace provides

the opportunity to define a new linear operator whose domain is this subspace.

More precisely, if W is a T -invariant subspace of V , then

the restriction T_W of T to W is a linear operator on W .

The following theorem illustrates

how the two operators $T : V \rightarrow V$ and $T_W : W \rightarrow W$ are linked.

Theorem 5.21. : Let $T : V \rightarrow V$ be a linear operator of an n -dimensional vector space V .

If W is a T -invariant subspace of V , then

the characteristic polynomial of $T_W : W \rightarrow W$ divides the characteristic polynomial of $T : V \rightarrow V$.

Theorem 5.21. : Let $T : V \rightarrow V$ be a linear operator of an n -dimensional vector space V .

If W is a T -invariant subspace of V , then

the characteristic polynomial of $T_W: W \rightarrow W$ divides the characteristic polynomial of $T: V \rightarrow V$.

Proof : Let $\gamma = \{w_1, \dots, w_k\}$ be an ordered basis for W .

We can extend γ to an ordered basis $\beta = \{w_1, \dots, w_k, v_{k+1}, \dots, v_n\}$ for V .

If $A := [T]_\beta$ and $B_1 := [T_W]_\gamma$, then we have $A = \begin{pmatrix} B_1 & B_2 \\ O & B_3 \end{pmatrix}$.

Then

$$\begin{aligned} \text{the characteristic polynomial of } T &= \det(A - tI_n) = \det \begin{pmatrix} B_1 - tI_k & B_2 \\ O & B_3 - tI_{n-k} \end{pmatrix} \\ &= \det(B_1 - tI_k) \times \det(B_3 - tI_{n-k}). \end{aligned}$$

This completes the proof since

$\det(B_1 - tI_k)$ is the characteristic polynomial of $T_W: W \rightarrow W$.



Lemma : Let B and D be respectively $k \times k$ and $(n - k) \times (n - k)$ matrix.

$$\text{If } A = \begin{pmatrix} B & C \\ O & D \end{pmatrix}, \text{ then } \det(A) = \det(B) \times \det(D).$$

Proof : We use induction on $k \geq 1$.

If $k = 1$, then it is obvious that $\det(A) = \det(B_1) \times \det(B_2)$.

Suppose that $k \geq 2$. Then

$$\det(A) = \sum_{i=1}^k (-1)^{i+1} a_{i1} \det(A_{i1}).$$

Also, by induction hypothesis, we have

$$\det(A_{i1}) = \det(B_{i1}) \det(D) \text{ for all } 1 \leq i \leq k.$$

Therefore, it follows that

$$\det(A) = \left(\sum_{i=1}^k (-1)^{i+1} a_{i1} \det(B_{i1}) \right) \times \det(D) = \det(B) \times \det(D).$$

Example : For $V = P_n(\mathbb{R})$ and $\beta = \{1, x, x^2, \dots, x^n\}$, consider the linear transformations

$$T_1 : V \rightarrow V, \quad f(x) \mapsto f'(x),$$

and

$$T_2 : V \rightarrow V, \quad f(x) \mapsto xf'(x).$$

$$(1) [T_1]_{\beta} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & n \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}$$

Hence the characteristic polynomial of T_1 is equal to $(-1)^n t^n$.

$$(2) [T_2]_{\beta} = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & n \end{pmatrix}$$

Hence the characteristic polynomial of T_1 is equal to $(-1)^n t(t-1)(t-2) \cdots (t-n)$.

Example (Continued) :

$$T_1 : V \rightarrow V, \quad f(x) \mapsto f'(x), \text{ and } T_2 : V \rightarrow V, \quad f(x) \mapsto xf'(x)$$

Let $f(x) = x^2 + x + 1$.

(3) The T_1 -cyclic subspace of V generated by $f(x)$ is

$$\begin{aligned} W &= \text{span}(\{f(x), T_1(f(x)), T_1^2(f(x)), \dots\}) = \text{span}(\{x^2 + 2x + 1, 2x + 1, 2, 0, \dots\}) \\ &= \text{span}(\{1, x, x^2\}) = P_2(\mathbb{R}). \end{aligned}$$

For the standard basis $\gamma = \{1, x, x^2\}$ of $P_2(\mathbb{R})$, we have

$$[T_1|_W]_\gamma = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

and so

the characteristic polynomial of $T_1|_W$ is equal to $-t^3$.

Example (Continued) :

$$T_1 : V \rightarrow V, \quad f(x) \mapsto f'(x), \text{ and } T_2 : V \rightarrow V, \quad f(x) \mapsto xf'(x)$$

Let $f(x) = x^2 + x + 1$.

(4) The T_2 -cyclic subspace of V generated by $f(x)$ is

$$\begin{aligned} W &= \text{span}(\{f(x), T_2(f(x)), T_2^2(f(x)), \dots\}) = \text{span}(\{x^2 + 2x + 1, 2x^2 + x, 4x^2 + x, \dots\}) \\ &= \text{span}(\{1, x, x^2\}) = P_2(\mathbb{R}). \end{aligned}$$

For the standard basis $\gamma = \{1, x, x^2\}$ of $P_2(\mathbb{R})$, we have

$$[T_2|_W]_\gamma = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

and so

the characteristic polynomial of $T_2|_W$ is equal to $-t(t-1)(t-2)$.

Example 5. : Let T be the linear operator on $\mathbb{R}^4$ defined by

$$T(a,b,c,d) = (a+b+2c-d, b+d, 2c-d, c+d).$$

(1) Prove that $W = \{ (t,s,0,0) \mid t,s \in \mathbb{R} \}$ is a T -invariant subspace of $\mathbb{R}^4$.

($\because$) One can directly show that $T(w) \subset W$ for all $w \in W$.

For another proof, choose $v = (0,1,0,0)$.

Then we have $T(v) = (1,1,0,0)$ and $T^2(v) = (2,2,0,0) = 2T(v)$.

Since $T^2(v) \in \text{span}(v, T(v))$, it follows that

the T -cyclic subspaces of $\mathbb{R}^4$ generated by v is equal to $\text{span}(v, T(v))$.

It can be easily shown that $\text{span}(v, T(v))$ is equal to W .



(2) For $\gamma = \{e_1, e_2\}$, we have $B_1 := [T_W]_\gamma = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Then Theorem 5.21 says that

$\det(B_1 - tI) = \det\begin{pmatrix} 1-t & 1 \\ 0 & 1-t \end{pmatrix} = t^2 - 2t + 1 = (t-1)^2$ divides the characteristic polynomial of T .

Example 5. (Continued) : Let $\beta = \{e_1, e_2, e_3, e_4\}$. Then

$$A = [T]_{\beta} = \begin{pmatrix} 1 & 1 & 2 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

and hence the characteristic polynomial of T is equal to

$$\begin{aligned} f(t) &= \det(A - tI_4) = \det \begin{pmatrix} 1-t & 1 & 2 & -1 \\ 0 & 1-t & 0 & 1 \\ 0 & 0 & 2-t & -1 \\ 0 & 0 & 1 & 1-t \end{pmatrix} \\ &= \det \begin{pmatrix} 1-t & 1 \\ 0 & 1-t \end{pmatrix} \cdot \det \begin{pmatrix} 2-t & -1 \\ 1 & 1-t \end{pmatrix} \\ &= (t-1)^2 \times (t^2 - 3t + 3). \end{aligned}$$

Remark : In view of Theorem 5.21, we may use the characteristic polynomial of T_W to gain information about the characteristic polynomial of T itself.

In this regard, the next theorem explains how to compute

the characteristic polynomial of the restriction of a linear operator T to a cyclic subspace.

Theorem 5.22 : Let $T : V \rightarrow V$ be a linear operator on a finite dimensional vector space V .

For a nonzero vector $v \in V$, let

$$W = \text{span}(\{v, T(v), T^2(v), \dots\}) = \text{span}(\{T^\ell(v) \mid \ell \geq 0\})$$

denote the T -cyclic subspace of V generated by v . If $\dim W$ is k , then

(1) $\{v, T(v), \dots, T^{k-1}(v)\}$ is a basis for W .

(2) If $a_0v + a_1T(v) + \dots + a_{k-1}T^{k-1}(v) + T^k(v) = O_V$ for some $a_0, a_1, \dots, a_{k-1} \in F$, then

the characteristic polynomial of T_W is $f(t) = (-1)^k(a_0 + a_1t + \dots + a_{k-1}t^{k-1} + t^k)$.

Proof of (1) : Let j be the largest positive integer such that

$$\beta := \{v, T(v), \dots, T^{j-1}(v)\} \text{ is linearly independent.}$$

Such a j must exist since $v \neq O_V$ and $\dim V$ is finite.

Then $\{v, T(v), \dots, T^{j-1}(v), T^j(v)\}$ is linearly dependent and so

$$a_0v + a_1T(v) + \dots + a_{j-1}T^{j-1}(v) + a_jT^j(v) = O_V \text{ for some } (a_0, a_1, \dots, a_j) \in F^{j+1} - \{(0, \dots, 0)\}.$$

Note that $a_j \neq 0$ since otherwise $a_0, \dots, a_{j-1}$ are all zeros.

Thus we may assume that $a_j = 1$ and so

$$a_0v + a_1T(v) + \dots + a_{j-1}T^{j-1}(v) + T^j(v) = O_V.$$

Then $T^j(v) = -a_0v - a_1T(v) - \dots - a_{j-1}T^{j-1}(v)$ is contained in $\text{span}(\beta)$.

Thus $W = \text{span}(\beta)$ and hence $\dim W = k = |\beta| = j$.

This proves that $j = k$ and so $\{v, T(v), \dots, T^{k-1}(v)\}$ is a basis for W .

Proof of (2) : From now on, we view $\beta = \{v, T(v), \dots, T^{j-1}(v)\}$ as an ordered basis for W .

Let $a_0, a_1, \dots, a_{k-1} \in F$ be as above so that $a_0v + a_1T(v) + \dots + a_{j-1}T^{k-1}(v) + T^k(v) = O_V$.

Then

$$\begin{aligned} [T_W]_\beta &= [[T_W(v)]_\beta \ [T_W(T(v))]_\beta \cdots [T_W(T^{j-2}(v))]_\beta \ [T_W(T^{j-1}(v))]_\beta] \\ &= [[T(v)]_\beta \ [T^2(v)]_\beta \cdots [T^{j-1}(v)]_\beta \ [T^j(v)]_\beta] \\ &= \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{k-1} \end{pmatrix}. \end{aligned}$$

Thus the characteristic polynomial of $[T_W]_\beta$ is

$$f(t) = \det \begin{pmatrix} -t & 0 & \cdots & 0 & -a_0 \\ 1 & -t & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -t - a_{k-1} \end{pmatrix}.$$

Proof of (2) - Continued : Claim (★) : $f(t) = (-1)^k(a_0 + a_1t + \cdots + a_{k-1}t^{k-1} + t^k).$

To prove Claim (★), we use induction on k .

- If $k = 1$, then $f(t) = \det(-t - a_0) = (-1)^1(t + a_0).$
- If $k = 2$, then $f(t) = \det\begin{pmatrix} -t & -a_0 \\ 1 & -t - a_1 \end{pmatrix} = t^2 + a_1t + a_0.$

Now, suppose that $k \geq 3$ and Claim (★) holds for any $\ell \leq k-1$. Then

$$\begin{aligned} f(t) &= \det \begin{pmatrix} -t & 0 & \cdots & 0 & -a_0 \\ 1 & -t & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -t - a_{k-1} \end{pmatrix} \\ &= (-t) \times \det \begin{pmatrix} -t & 0 & \cdots & 0 & -a_1 \\ 1 & -t & \cdots & 0 & -a_2 \\ 0 & 1 & \cdots & 0 & -a_3 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -t - a_{k-1} \end{pmatrix} + (-1)^{1+k}(-a_0) \det \begin{pmatrix} 1 & -t & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \end{aligned}$$

$$\begin{aligned}
&= (-t)(-1)^{k-1}(a_1 + a_2t + \cdots + a_{k-1}t^{k-2} + t^{k-1}) + (-1)^k \times a_0 \\
&= (-1)^k(a_0 + a_1t + \cdots + a_{k-1}t^{k-1} + t^k).
\end{aligned}$$

■

Theorem 5.23 (Cayley-Hamilton) : For a linear operator T on an n -dimensional vector space V , let $f(t) = b_0 + b_1t + \cdots + b_nt^n$ be its characteristic polynomial. Then T satisfies $f(t)$ in the sense that

$$f(T) := b_0Id_V + b_1T + b_2T^2 + \cdots + b_nT^n \text{ is the zero map.}$$

Proof : We need to show that $f(T)(v) = O_V$ for all $v \in V$.

This is obvious if $v = O_V$. So, we assume that $v \neq O_V$.

Let W be the T -cyclic subspace generated by v , and suppose that $\dim W = k$.

Let $g(t) = (-1)^k(a_0 + a_1t + \cdots + a_{k-1}t^{k-1} + t^k)$ be the characteristic polynomial of T_W .

Then **Theorem 5.22** says that $g(T)(v) = O_V$.

Also, by **Theorem 5.21**, we have $f(t) = q(t) \times g(t)$ for some polynomial $q(t)$.

Thus $f(T) = q(T)g(T)$ and hence

$$f(T)(v) = q(T)(g(T)(v)) = q(T)(O_V) = O_V.$$

■

Corollary (Cayley-Hamilton Theorem for Matrices) :

For $A \in M_{n \times n}(F)$, let $f(t)$ be the characteristic polynomial of A .

Then $f(A) = O_{n \times n}$ where $O_{n \times n}$ denotes the $n \times n$ zero matrix.

Example 7. : Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be such that $T(a, b) = (a - 2b, -2a + b)$.

For $\beta = \{e_1, e_2\}$, we have

$$A := [T]_{\beta} = \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix}.$$

Thus the characteristic polynomial of T is $\det(A - tI) = \det \begin{pmatrix} 1-t & -2 \\ -2 & 1-t \end{pmatrix} = t^2 - 2t - 3$.

Thus, by Cayley-Hamilton, we have $A^2 - 2A - 3I = O$.

Now, find A^{2024} .

$$- t^{2024} = (t^2 - 2t - 3)q(t) + at + b = (t - 3)(t + 1)q(t) + at + b :$$

If $t = -1$, then $-a + b = 1$. Also if $t = 3$, then $3a + b = 3^{2024}$.

$$\text{Thus } a = \frac{3^{2024} - 1}{4} \text{ and } b = \frac{3^{2024} + 3}{4}$$

$$- A^{2024} = aA + bI = \frac{3^{2024} - 1}{4} \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix} + \frac{3^{2024} + 3}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{3^{2024}+1}{2} & \frac{-3^{2024}+1}{2} \\ \frac{-3^{2024}+1}{2} & \frac{3^{2024}+1}{2} \end{pmatrix}.$$

Example 8. : Let $T : P_2(\mathbb{R}) \rightarrow P_2(\mathbb{R})$ be such that $T(f(x)) = f(0)x^2 + f'(1)x + f''(-1)$.

For $\beta = \{1, x, x^2\}$, let $A = [T]_\beta$. Then find A^{10} .

$$(1) \quad T(1) = x^2, \quad T(x) = x, \quad T(x^2) = 2x + 2$$

$$\text{Thus } A = [T]_\beta = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 0 & 0 \end{pmatrix} \text{ and hence}$$

$$\det(A - tI) = \det \begin{pmatrix} -t & 0 & 2 \\ 0 & 1-t & 2 \\ 1 & 0 & -t \end{pmatrix} = -t(t^2 - t) + 2(t - 1) = -(t^3 - t^2 - 2t + 2).$$

(2) By **Cayley-Hamilton Theorem**, we have $A^3 - A^2 - 2A + 2I_3 = O_{3 \times 3}$.

Thus $A^3 = A^2 + 2A - 2I_3$. Then

$$A^4 = A^3 + 2A^2 - 2A = 3A^2 - 2I_3 \text{ and } A^5 = 3A^3 - 2A = 3A^2 + 4A - 6I_3.$$

$$\begin{aligned}
\text{Then } A^{10} &= 9A^4 + 16A^2 + 36I_3 + 24A^3 - 36A^2 - 48A \\
&= 9(3A^2 - 2I_3) + 16A^2 + 36I_3 + 24(A^2 + 2A - 2I_3) - 36A^2 - 48A \\
&= 31A^2 - 30I_3.
\end{aligned}$$

Example 8. (Continued) :

(3) Since $A = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 0 & 0 \end{pmatrix}$ and so

$$A^2 = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 2 & 1 & 2 \\ 0 & 0 & 2 \end{pmatrix},$$

it follows that

$$\begin{aligned}
A^{10} &= 31A^2 - 30I_3 \\
&= 31 \begin{pmatrix} 2 & 0 & 0 \\ 2 & 1 & 2 \\ 0 & 0 & 2 \end{pmatrix} - 30 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 32 & 0 & 0 \\ 62 & 1 & 62 \\ 0 & 0 & 32 \end{pmatrix}.
\end{aligned}$$

[A의 Diagonalization을 이용하는 방법]

$$f(t) = \det(A - tI) = \det \begin{pmatrix} -t & 0 & 2 \\ 0 & 1-t & 2 \\ 1 & 0 & -t \end{pmatrix} = t^2(1-t) - 2(1-t) = (1-t)(t^2 - 2)$$

$$t = 1 : \begin{pmatrix} -1 & 0 & 2 \\ 0 & 0 & 2 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow N(A - I) = \left\{ y \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \mid y \in \mathbb{R} \right\}$$

$$t = \sqrt{2} : \begin{pmatrix} -\sqrt{2} & 0 & 2 \\ 0 & 1-\sqrt{2} & 2 \\ 1 & 0 & -\sqrt{2} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow N(A - \sqrt{2}I) = \left\{ z \begin{pmatrix} \sqrt{2} \\ 2+2\sqrt{2} \\ 1 \end{pmatrix} \mid z \in \mathbb{R} \right\}$$

$$t = -\sqrt{2} : \begin{pmatrix} \sqrt{2} & 0 & 2 \\ 0 & 1+\sqrt{2} & 2 \\ 1 & 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow N(A + \sqrt{2}I) = \left\{ z \begin{pmatrix} -\sqrt{2} \\ 2-2\sqrt{2} \\ 1 \end{pmatrix} \mid z \in \mathbb{R} \right\}$$

$$\Rightarrow \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & \sqrt{2} & -\sqrt{2} \\ 1 & 2+2\sqrt{2} & 2-2\sqrt{2} \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & \sqrt{2} & -\sqrt{2} \\ 1 & 2+2\sqrt{2} & 2-2\sqrt{2} \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & -\sqrt{2} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \sqrt{2} & -\sqrt{2} \\ 1 & 2+2\sqrt{2} & 2-2\sqrt{2} \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & -\sqrt{2} \end{pmatrix} \begin{pmatrix} 0 & \sqrt{2} & -\sqrt{2} \\ 1 & 2+2\sqrt{2} & 2-2\sqrt{2} \\ 0 & 1 & 1 \end{pmatrix}^{-1}$$

Since $\begin{pmatrix} 0 & \sqrt{2} & -\sqrt{2} \\ 1 & 2+2\sqrt{2} & 2-2\sqrt{2} \\ 0 & 1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} -2 & 1 & 2 \\ \frac{\sqrt{2}}{4} & 0 & \frac{1}{2} \\ -\frac{\sqrt{2}}{4} & 0 & \frac{1}{2} \end{pmatrix}$, we have

$$\begin{aligned} \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 0 & 0 \end{pmatrix}^n &= \begin{pmatrix} 0 & \sqrt{2} & -\sqrt{2} \\ 1 & 2+2\sqrt{2} & 2-2\sqrt{2} \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \sqrt{2}^n & 0 \\ 0 & 0 & (-1)^n \sqrt{2}^n \end{pmatrix} \begin{pmatrix} -2 & 1 & 2 \\ \frac{\sqrt{2}}{4} & 0 & \frac{1}{2} \\ -\frac{\sqrt{2}}{4} & 0 & \frac{1}{2} \end{pmatrix} \\ &= \begin{pmatrix} 0 & \sqrt{2} & -\sqrt{2} \\ 1 & 2+2\sqrt{2} & 2-2\sqrt{2} \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} -2 & 1 & -2 \\ \sqrt{2}^{n-3} & 0 & \sqrt{2}^{n-2} \\ (-1)^{n+1} \sqrt{2}^{n-3} & 0 & (-1)^n \sqrt{2}^{n-2} \end{pmatrix} \end{aligned}$$

$$= \begin{pmatrix} \delta_n \sqrt{2^{n-2}} & 0 & \delta_{n+1} \sqrt{2^{n-1}} \\ -2 + \delta_{n+1} \sqrt{2^{n-1}} + \delta_n \sqrt{2^n} & 1 & -2 + \delta_{n+1} \sqrt{2^{n+1}} + \delta_n \sqrt{2^n} \\ \delta_{n+1} \sqrt{2^{n-3}} & 0 & \delta_n \sqrt{2^{n-2}} \end{pmatrix}$$

$$\text{where } \delta_k := 1 + (-1)^k.$$

$$\text{For example, } \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 0 & 0 \end{pmatrix}^9 = \begin{pmatrix} 0 & 0 & 16 \\ 30 & 1 & 62 \\ 16 & 0 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 0 & 0 \end{pmatrix}^{10} = \begin{pmatrix} 32 & 0 & 0 \\ 62 & 1 & 62 \\ 0 & 0 & 32 \end{pmatrix}.$$